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FMN040 Numerical Analysis for E3 Solutions for Homework assignment 6

1. Turbo Newton.

Task 1a): The implementation of the method is done in the following code

```
function y=turbonewton(fun,funx,funxx,x0,tol,exactval)

dx=30000;
iter=0;
x=x0;
while abs(dx/x)>tol & iter<10
    valxx=feval(funxx,x);
    valx=feval(funx,x);
    val=feval(fun,x);
    if ( double(valxx) == 0 ) | (double(valx^2)<double(2*valxx*val))
        dx=-val/valx;
    else
        dx=(-valx+sign(valx)*sqrt(valx^2-2*valxx*val))/valxx;
    end
    x=x+dx;
    fprintf('x:%1.15d\n',x);
    iter=iter+1;
    fprintf('Iteration: %d\n',iter);
    fprintf('Error: %1.15d\n',abs(x-exactval));
end

y=x;
```

Task 1b): The function $f(x) = e^x - \pi$ has derivatives $f'(x) = e^x$ and $f''(x) = e^x$. We write all this in *MATLAB* and obtain the following output

```
>> function=inline('exp(x)-pi','x');
>> funcionx=inline('exp(x)','x');
>> funcionxx=inline('exp(x)','x');
x=turbonewton(function,funcionx,funcionxx,1.096629,1e-16,log(pi));
x:1.144747797786381e+000
Iteration: 1
Error: 1.791193698030824e-005
x:1.144729885849399e+000
Iteration: 2
```

```
Error: 1.110223024625157e-015
x:1.144729885849400e+000
>>
```

2. Tunnel Diode Oscillator.

Task 2a): Introducing an auxiliar variable $u(t) = \dot{V}(t)$ the system can be written as a first order differential equation as follows

$$\dot{Y}(t) = \begin{pmatrix} \dot{V}(t) \\ \dot{u}(t) \end{pmatrix} = F(t, V(t), u(t))$$

with

$$F(t, V(t), u(t)) = \begin{pmatrix} \alpha u(t) - 3\beta V^2(t)u(t) - \frac{1}{LC}V(t) \\ u(t) \end{pmatrix}.$$

The function F is the right hand side needed for simulation and it is implemented in the following code

```
function y=vanderpol(t,Y);
alpha=1e4;
beta=1e6;
L=1e-5;
C=1e-5;

omega=sqrt(1/L/C);
y=[alpha*Y(1)-3*beta*Y(2)^2*Y(1)-omega^2*Y(2); Y(1)];
```

Task 2b): A general code using the function `feval` is the following

```
function [t,y,e]=eulerexpimp(funcion,t0,tf,n,y0)

t=linspace(t0,tf,n);
y=zeros(size(y0,1),n);
h=t(2)-t(1);
e=zeros(size(y0,1),n);

y(:,1)=y0;

for i=2:n
    temp=y(:,i-1)+h*feval(funcion,t(i-1),y(:,i-1));
    y(:,i)=y(:,i-1)+h*feval(funcion,t(i),temp);
    for k=1:2
```

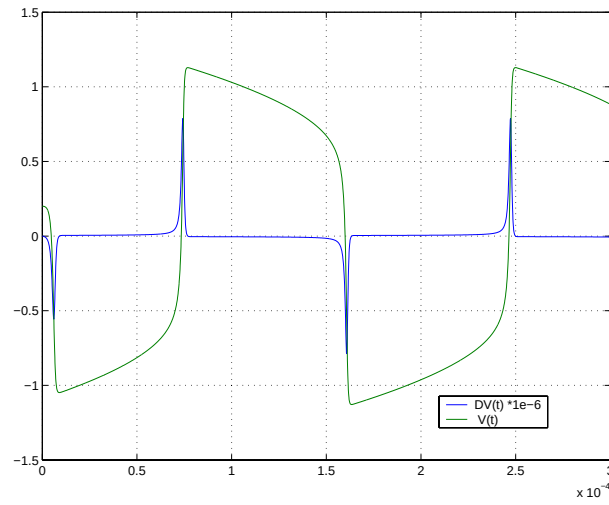


Figure 1: Solution for $\alpha = 10^6$

```

        y(:,i)=y(:,i-1)+h*feval(funcion,t(i),y(:,i));
    end
    e(:,i)=temp-y(:,i);
end

```

The approximated solutions are depicted in Fig. 1 and Fig. 2. Notice that one solution is smooth and the other has discontinuities. This factor will play a major role when we try to control the step size.

Task 2c): The figures show the estimated error and the functions $u(t)$ and $v(t)$. Notice the factors and how the error grows dramatically in the neighborhood of the discontinuities and decrease were the functions are continuous and resembles a straight line.

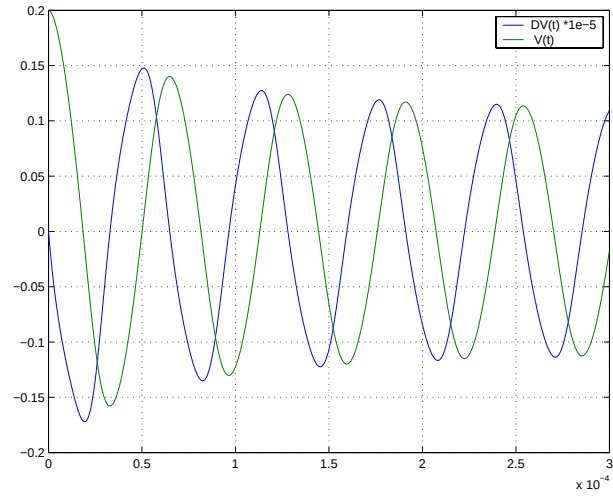


Figure 2: Solution for $\alpha = 10^4$

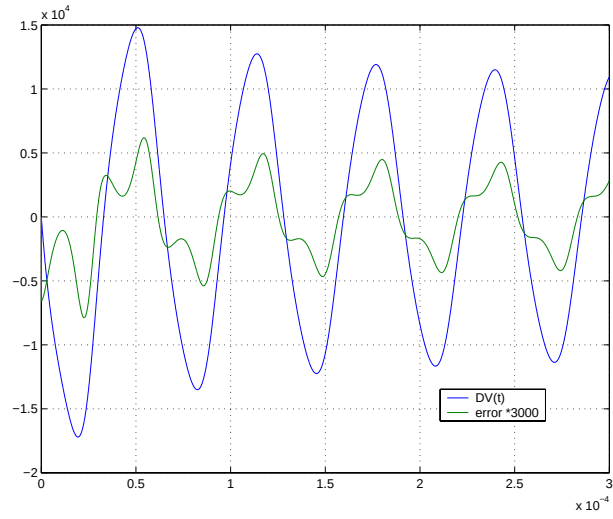


Figure 3: $u(t)$ and estimated global error for $\alpha = 10^4$

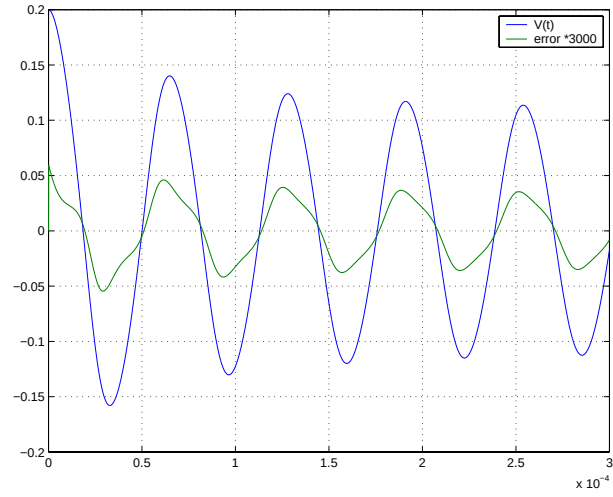


Figure 4: $v(t)$ and estimated global error for $\alpha = 10^4$

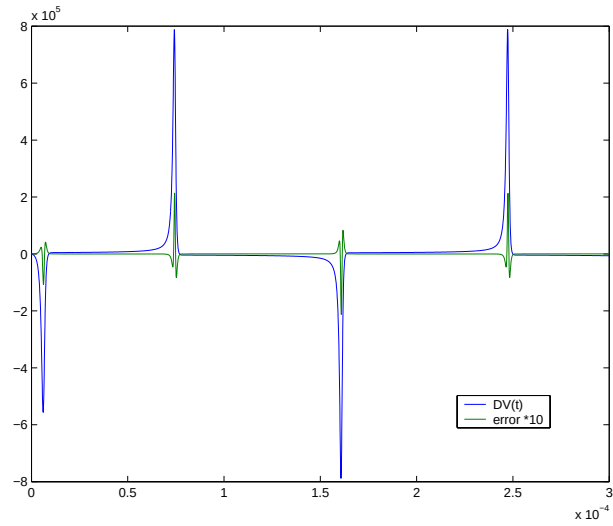


Figure 5: $u(t)$ and estimated global error for $\alpha = 10^6$

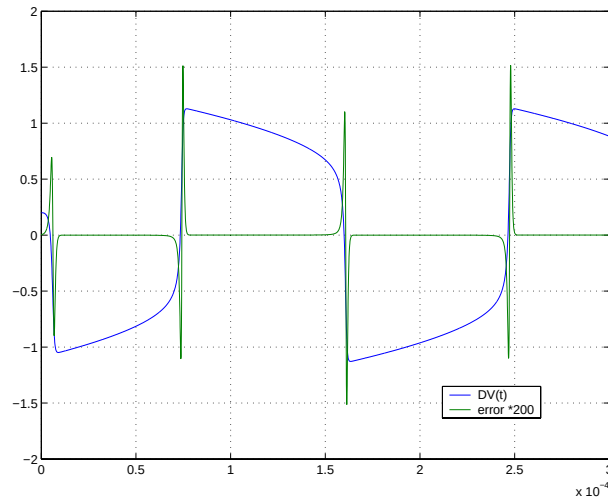


Figure 6: $v(t)$ and estimated global error for $\alpha = 10^6$